

Disaster Management System

By Remotz Systems

Deliverable-3

Project Team

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Semester 6

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PART B — Development Methodology (Scrum) - Sprint 3

1. Sprint Backlog for Sprint 3

1.1 Sprint Duration: March 31, 2026 - April 18, 2026 (2.75 weeks)

1.2 Sprint Goal: Complete the disaster response coordination setup by implementing NGO/Rescue Team assignment, real-time mission tracking, status management, report generation, and feedback collection, delivering the fully integrated Disaster Management System.

1.3 Modules Developed in Sprint 3: Coordination, Monitoring & System Completion

- Assign NGO (US-19 to US-21)
- Assign Rescue Team (US-22 to US-24)
- Track Rescue Team (US-25 to US-27)
- Update Mission Status (US-28 to US-30)
- Generate Disaster Report (US-31 to US-33)
- Give Feedback & Review Feedback (US-34, US-35)

2. Sprint Backlog for Sprint 3

2.1 Module Name

Coordination, Monitoring & System Completion

2.2 User Stories

User Story 19:

Story ID: US-19	Story Title: Assign NGO	
User Story		Importance:
As an Admin, I want to assign NGOs to disaster-affected locations so that I can dispatch an appropriate NGO efficiently.		High
		Estimate:
		3 day
Acceptance Criteria		Type:
• The system provides options to track the availability of NGOs • System validates that only available NGOs are assigned. • If an unavailable NGO is assigned, the system shows an error message. • System updates the NGO database after confirmation. • System logs all NGO assignment actions.		☐ Search ☒ Workflow ☐ Manage Data ☐ Payment ☐ Report / View

User Story 20:

Story ID: US-20	Story Title: View Available NGOs	
User Story		Importance:
		Medium

As an Admin, I want to be able to view all available NGOs so I can make decisions on which NGO is to be assigned to a specific mission.	Estimate:
	1 day
Acceptance Criteria	Type:
- The system provides a tabular display listing NGOs, their information(location, personnel, speciality) and availability. - The system updates the status of availability as soon as the NGO is assigned.	☐ Search ☐ Workflow ☐ Manage Data ☐ Payment ☒ Report / View

User Story 21:

Story ID: US-21	Story Title: Assign Rescue Team	
User Story	Importance:	
As an Admin, I want to assign Rescue Teams to disaster-affected locations so that I can dispatch an appropriate Rescue Team efficiently.	High	
	Estimate:	
	3 day	
Acceptance Criteria	Type:	
- The system provides options to track the availability of Rescue Teams - System validates that only available Rescue Teams are assigned. - If an unavailable Rescue Team is assigned, the system shows an error message. - System updates the Rescue Team database after confirmation. - System logs all Rescue Team assignment actions.	☐ Search ☒ Workflow ☐ Manage Data ☐ Payment ☐ Report / View	

User Story 23:

Story ID: US-23	Story Title: View Available Rescue Teams	
User Story	Importance:	
As an Admin, I want to view available rescue teams so that I can select the most suitable team for disaster response.	Medium	
	Estimate:	
	2 days	
Acceptance Criteria	Type:	
- System displays a list of all rescue teams. - System shows team details (location, availability, specialization).	☐ Search ☐ Workflow ☐ Manage Data	

• System filters only available teams. • If no teams are available, system shows an appropriate message. • Data is retrieved in real-time.	☐ Payment ☒ Report / View
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User Story 24:

Story ID: US-24	Story Title: Assign Team to Disaster		
User Story		Importance:	
As an Admin, I want to assign a rescue team to a disaster so that rescue operations can start quickly and efficiently.		High	
		Estimate:	
		3 days	
Acceptance Criteria		Type:	
• System displays disasters requiring rescue teams. • System shows available rescue teams. • Admin selects a team for a disaster. • System validates availability of selected team. • Upon confirmation, system assigns the team. • System notifies the rescue team. • System updates disaster and team status.		☐ Search ☒ Workflow ☒ Manage Data ☐ Payment ☐ Report / View	

User Story 25:

Story ID: US-25	Story Title: Track Rescue Team		
User Story		Importance:	
As an Admin, I want to track rescue teams so that I can monitor their progress and ensure timely response.		High	
		Estimate:	
		4 days	
Acceptance Criteria		Type:	
• System provides tracking option for rescue teams. • System retrieves real-time team data. • Admin can select a specific team to track. • System displays current mission details. • If team is offline, system shows alert message.		☐ Search ☐ Workflow ☐ Manage Data ☐ Payment ☒ Report / View	

User Story 26:

Story ID: US-26	Story Title: View Active Teams Dashboard	
User Story	Importance:	
As an Admin, I want to view an active teams dashboard so that I can get an overview of all ongoing rescue operations.	Medium	
	Estimate:	
	2 days	
Acceptance Criteria	Type:	
- System displays all active rescue teams. - Dashboard shows assigned disasters and status. - Data updates in real-time. - If no teams are active, system displays a message.	☐ Search ☐ Workflow ☐ Manage Data ☐ Payment ☒ Report / View	

User Story 27:

Story ID: US-27	Story Title: View Team Location & Status	
User Story	Importance:	
As an Admin, I want to view team location and status so that I can monitor progress and coordinate effectively.	High	
	Estimate:	
	2 days	
Acceptance Criteria	Type:	
- System shows real-time location of selected team. - System displays mission status (assigned, in progress, completed). - System updates data dynamically. - If location is unavailable, system shows an alert.	☐ Search ☐ Workflow ☐ Manage Data ☐ Payment ☒ Report / View	

User Story 28:

Story ID: US-28	Story Title: Update Mission Status	
User Story	Importance:	
As an NGO or Rescue Team member, I want to update mission status so that the admin stays informed about progress.	High	
	Estimate:	
	3 days	
Acceptance Criteria	Type:	
System allows NGO/Rescue Team to update mission status.	☐ Search	

• System provides status options (In Progress, Completed). • System validates the update. • System records changes in database. • System notifies admin of updates.	■ Workflow ■ Manage Data ☐ Payment ☐ Report / View
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User Story 29:

Story ID: US-29	Story Title: Mark Mission In Progress	
User Story	Importance:	
As an Admin, I want to view available rescue teams so that I can select the most suitable team for disaster response.	Medium	
	Estimate:	
	1 days	
Acceptance Criteria	Type:	
• System allows selection of assigned mission. • User selects “In Progress” status. • System validates the action. • System updates mission status in database. • Admin is notified of the update.	☐ Search ■ Workflow ☐ Manage Data ☐ Payment ☐ Report / View	

User Story 30:

Story ID: US-30	Story Title: Mark Mission Completed	
User Story	Importance:	
As an Admin, I want to view available rescue teams so that I can select the most suitable team for disaster response.	Medium	
	Estimate:	
	1 days	
Acceptance Criteria	Type:	
• System allows selection of assigned mission. • User selects “Completed” status. • System validates completion. • System updates mission status in database. • Admin is notified of completion. • System logs mission completion.	☐ Search ■ Workflow ☐ Manage Data ☐ Payment ☐ Report / View	

User Story: 31

Story ID: US-31	Story Title: Generate Disaster Report
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User Story	Importance:
As an Admin, I want to generate detailed reports so that I can review disaster impact, resources used, and team efficiency.	High
	Estimate:
	34 days
Acceptance Criteria	Type:
• The system must compile disaster events, affected areas, response teams, relief resources distributed, and outcomes. • The system must log the report generation action for auditing purposes. • If report generation fails due to a system error, the admin must be prompted to retry.	☐ Search ☐ Workflow ☐ Manage Data ☐ Payment ☒ Report / View

User Story 32:

Story ID: US-32	Story Title: Set Report Parameters	
User Story	Importance:	
As an admin, I want to set specific parameters for the report so that the system can organize disaster data matching my criteria.	Medium	
	Estimate:	
	3 days	
Acceptance Criteria	Type:	
• System must prompt the admin for report parameters like date, disaster type, region, and teams involved. • Admin must be able to set desired parameters for the report into the system. • If no disaster match the entered parameters, the system must display the exact message: "No disaster data found for selected parameters".	☐ Search ☐ Workflow ☒ Manage Data ☐ Payment ☐ Report / View	

User Story 33:

Story ID: US-33	Story Title: Export/Download Report	
User Story	Importance:	
As an Admin, I want to select a report format so that I can view or download the generated report.	Medium	
	Estimate:	
	2 days	
Acceptance Criteria	Type:	
- System must prompt the admin to request a specific report format. - System must successfully generate the report in the chosen format, supporting options like PDF or Excel. - Admin must be able to view or download the report	☐ Search ☐ Workflow ☐ Manage Data ☐ Payment ☒ Report / View	

User Story 34:

Story ID: US-34	Story Title: Give Feedback	
User Story	Importance:	
As a citizen, NGO, or rescue team member, I want to share experiences or feedback for the system's performance so that the system could be improved.	Low	
	Estimate:	
	2 days	
Acceptance Criteria	Type:	
- System must provide a feedback form once the authenticated actor selects the option to give feedback. - If the feedback form is incomplete, the user must be prompted to fill in the mandatory fields. - If a database error occurs during submission, the system must show a failure message.	☐ Search ☒ Workflow ☐ Manage Data ☐ Payment ☐ Report / View	

User Story 35:

Story ID: US-35	Story Title: Review Feedback	
User Story	Importance:	
As an admin, I want to view the submitted feedback so that I can review the feedback for the system's performance.	Low	
	Estimate:	

	2 days
Acceptance Criteria	Type:
• System must store all recorded feedback properly. • All recorded feedback must be accessible for review by the admin through the interface.	☐ Search ☐ Workflow ☐ Manage Data ☐ Payment ☒ Report / View

2.3 Structured Specifications

Specification 9-US-19: Assign NGO, Rescue Team

Field	Details
Use Case Name	Assign NGO / Assign Rescue Team
Actor(s)	Admin
Inputs	• Admin selects an option to assign a team or NGO. • Selects a specific disaster. • Selects a relevant and available rescue team or NGO for the assignment.
Outputs	• System displays lists of disasters requiring assistance. • Shows available NGOs/Rescue Teams. • Notifies the assigned entity.

Processing	• System filters available teams based on location, availability, and skill • Filters NGOs based on relevance and availability • Assigns the team and updates disaster records and team status upon acknowledgement.
Preconditions	• Disaster events exist in the system • NGOs and rescue teams are registered and available.
Postconditions	• NGO or Rescue Team is assigned to the disaster incident • The assigned entity acknowledges the assignment.
Exceptions	• If no entities are available, the system alerts the admin • If admin cancels, the assignment is discarded • Admin can reschedule or reassign if the first rescue team cannot respond.

Specification 10-US-25: Track Rescue Team

Field	Details
Use Case Name	Track Rescue Team
Actor(s)	Admin

Inputs	• Admin selects options to track rescue teams • Selects a specific rescue team.
Outputs	• System retrieves a list of all active rescue teams and their assignments • Displays the team's current location, disaster assignment, status, and estimated arrival time.
Processing	System updates real-time location and status based on inputs from the rescue team.
Preconditions	• Rescue teams are registered in the system • Teams are assigned to at least one disaster.
Postconditions	Admin successfully views the real-time status of rescue teams.
Exceptions	• If a rescue team is offline/unreachable, the system alerts the admin • If no teams are active, the system displays "No active rescue teams currently".

Specification 11-US-31: Generate Disaster Report

Field	Details

Use Case Name	Generate Disaster Report
Actor(s)	Admin
Inputs	• Admin selects option to generate disaster reports • Enters report parameters (date, disaster type, region, teams involved) • Requests report format.
Outputs	• System prompts for parameters • Generates reports in the chosen format • Displays or downloads the report.
Processing	• System compiles disaster events, affected areas, response teams, relief resources distributed, and outcomes • Logs report generation for auditing purposes.
Preconditions	Disaster events, rescue assignments, and resource records exist in the system.
Postconditions	Admin successfully generates and views/downloads the disaster reports.
Exceptions	• If no disasters match the parameters, the system displays "No disaster data found for selected criteria."

	• If report generation fails due to system error, admin is prompted to retry.
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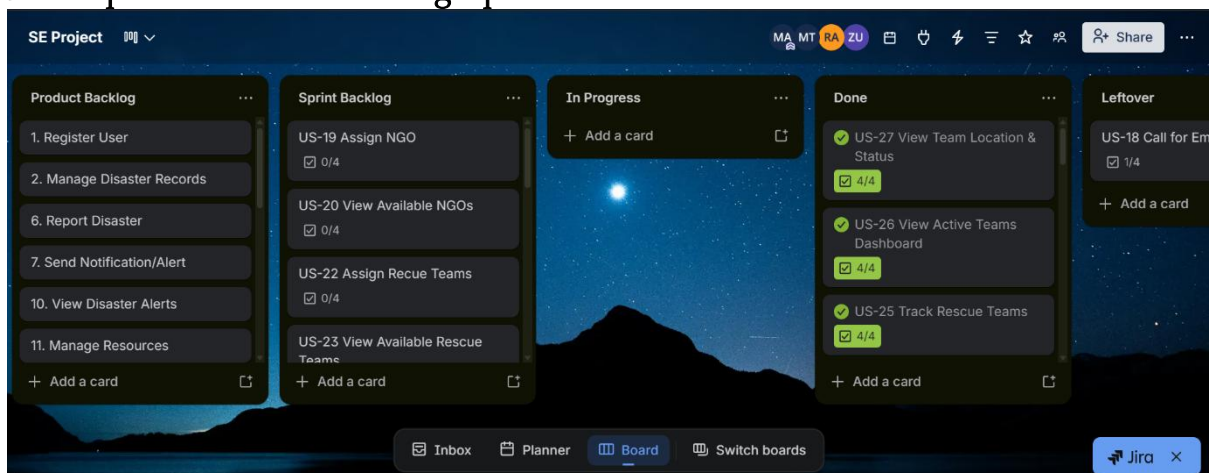
Specification 12-US-34: Give Feedback

Field	Details
Use Case Name	Give Feedback
Actor(s)	Citizen, NGO, Rescue Team
Inputs	• Actor selects the option to give feedback • Actor enters feedback into the provided form.
Outputs	System provides a feedback form.
Processing	• System records the submitted feedback • Admins can later review the feedback.

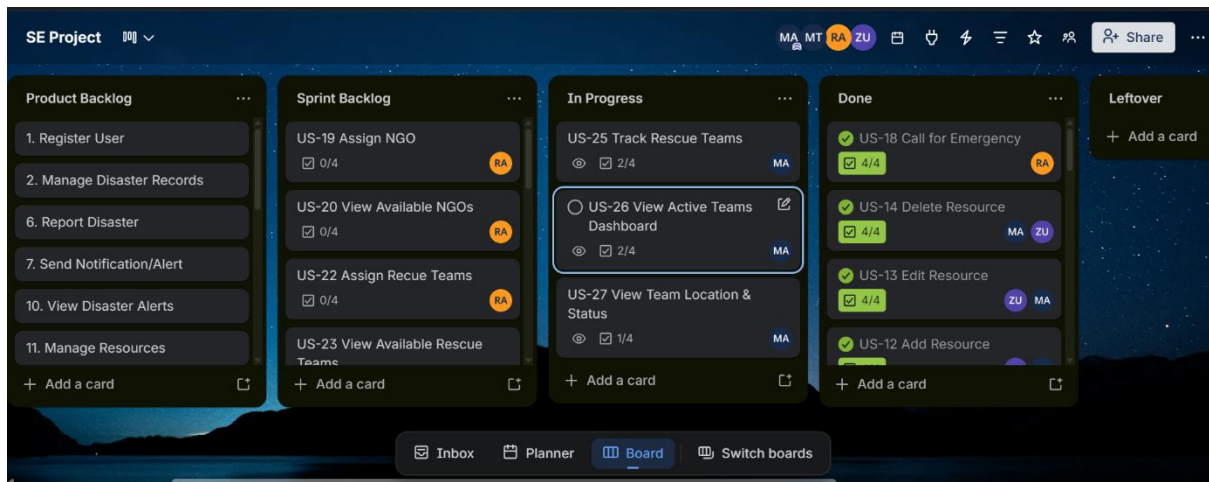
Preconditions	The actor is authenticated.
Postconditions	• Feedback is recorded • Respective feedback is accessible to admin, NGO, rescue teams.
Exceptions	• If the feedback form is incomplete, the user is prompted to fill mandatory fields • If a database error occurs, the system shows a failure message.

3. Scrum Board (Trello Snapshots)

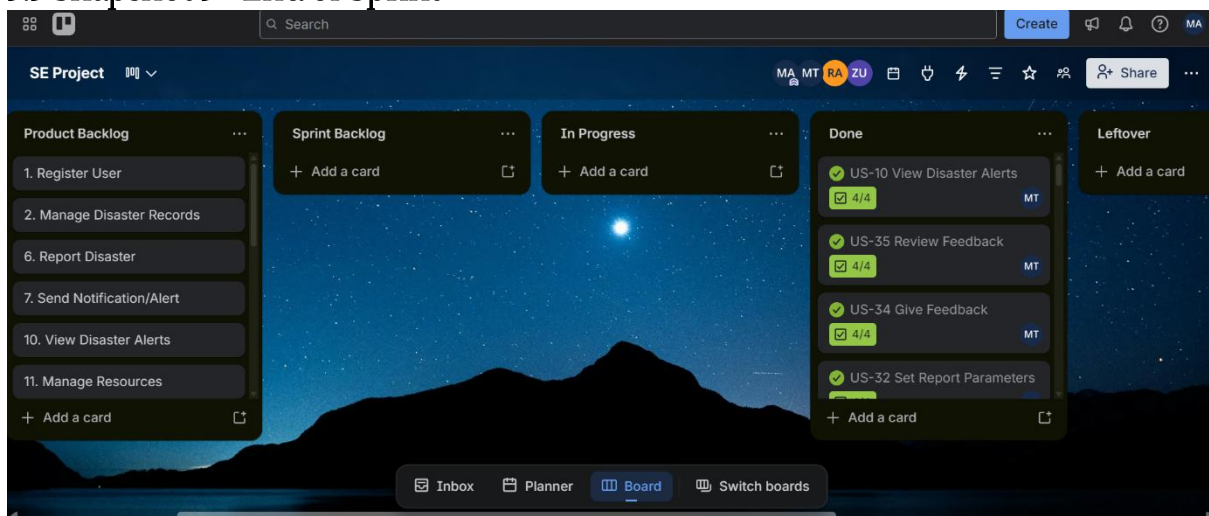
3.1 Snapshot 1 - Initial Backlog Upload



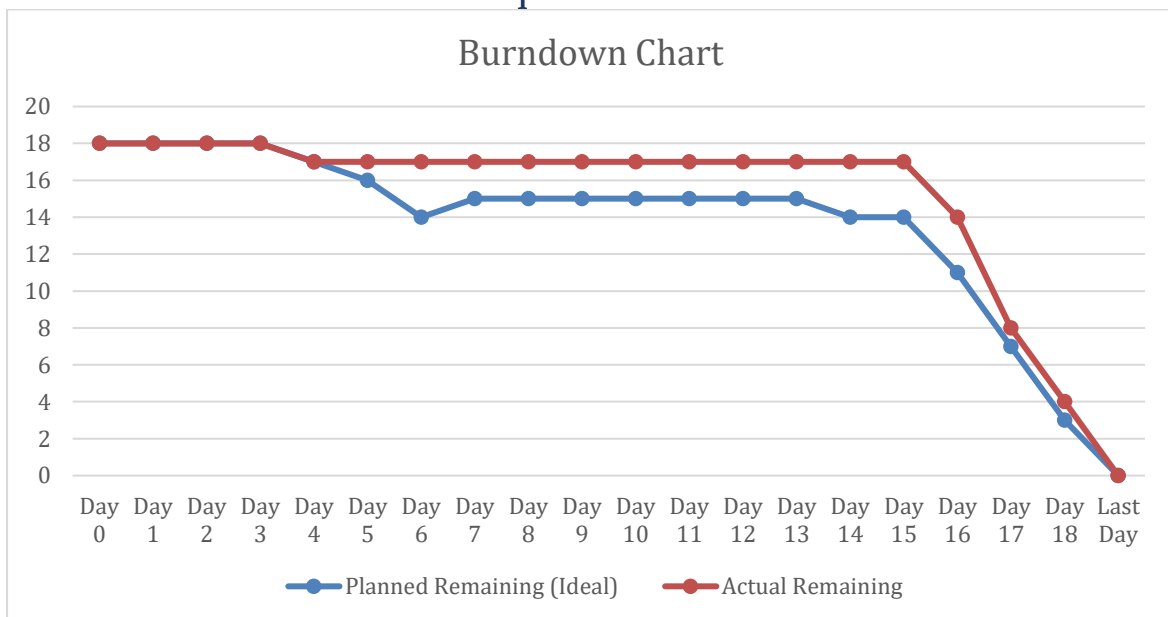
3.2 Snapshot 2 - Mid-Sprint Progress



3.3 Snapshot 3 - End of Sprint



4. Product Burn-Down Chart - Sprint 3



5. Testing Activities - Sprint 3

5.1. Software Test Plan

Test Plan Title	Sprint 3 Software Test Plan — Disaster Management System
Version	1.0
Prepared By	Remozone Systems — Muhammad Moiz Ansari, Rehan Tajik, Zohaib Umer, Muhammad Tayyab
Sprint	Sprint 3 — Coordination, Monitoring & System Completion
Test Period	April 1, 2026 – April 18, 2026
Scope	This test plan covers all Sprint 3 modules: • Assign NGO (US-19 to US-21) • Assign Rescue Team (US-22 to US-24) • Track Rescue Team (US-25 to US-27) • Update Mission Status (US-28 to US-30) • Generate Disaster Report (US-31 to US-33) • Give & Review Feedback (US-34, US-35)
Test Objectives	1. Verify that NGO and Rescue Team assignment correctly updates all related database records. 2. Verify that mission status transitions are enforced and reflected in real time. 3. Verify that report generation produces correct data for all filter combinations. 4. Verify that feedback submission and review workflows function correctly. 5. Verify that all exception paths display appropriate error messages without crashing.
Test Types	• Functional Testing — each use case against its acceptance criteria • Black-Box Testing — equivalence partitioning and boundary value analysis • White-Box Testing — statement, branch, and condition coverage on critical methods • Regression Testing — ensure Sprint 1 & Sprint 2 features remain unaffected
Tools & Environment	IDE: IntelliJ IDEA / Java 17 Database: SQL Server Express (local) Manual testing via JavaFX UI Test evidence: screenshots and console output logs
Entry Criteria	• Sprint 3 features are fully coded and compiled without errors. • Database schema matches the expected Sprint 3 design. • Sprint 1 & 2 features pass regression checks.

Exit Criteria	• All critical and high-priority test cases pass. • No open Critical or High severity defects remain unfixed. • All modules produce correct outputs for valid inputs. • Exception paths display user-friendly error messages.
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5.2. Black-Box Test Cases

Techniques used: Equivalence Partitioning (EP), Boundary Value Analysis (BVA), Error Guessing (EG)

TC-ID	Feature / Use Case	Input(s)	Expected Output	Technique	Result
TC-01	Assign NGO	Valid: Active disaster + Available NGO selected	NGO assigned; disaster record updated; success message	Equivalence Partitioning	Pass
TC-02	Assign NGO	Invalid: No NGOs available in system	Alert: 'No NGOs currently available.' Assignment blocked	Error Guessing	Pass
TC-03	Assign NGO	Invalid: Disaster already has NGO assigned	System prevents duplicate assignment	Equivalence Partitioning	Pass
TC-04	Assign Rescue Team	Valid: Active disaster + Available team selected	Team assigned; Mission record created; team status = Deployed	Equivalence Partitioning	Pass
TC-05	Assign Rescue Team	Invalid: Selected team has status 'Deployed'	Team card greyed out; cannot be selected	Equivalence Partitioning	Pass
TC-06	Track Rescue Team	Valid: Active missions exist in system	Teams list populated with correct status badges and ETAs	Equivalence Partitioning	Pass
TC-07	Track Rescue Team	Edge: No active missions	Empty state displayed: 'No active rescue teams currently.'	Boundary Value Analysis	Pass
TC-08	Update Mission Status	Valid: Change status from Assigned → On Site	Status updated in DB; timeline entry added; Admin dashboard refreshed	Equivalence Partitioning	Pass
TC-09	Update Mission Status	Edge: Status already Completed — attempt to change	System prevents change; status options for completed missions are disabled	Error Guessing	Pass
TC-10	Generate Report	Valid: Date range Jan 1 – Dec 31, All types, All severities	Report generated with correct totals; export buttons enabled	Equivalence Partitioning	Pass

TC-11	Generate Report	Invalid: Start date > End date	Error shown: 'Invalid date range.' Generation blocked	Boundary Value Analysis	Pass
TC-12	Generate Report	Edge: No disasters match the filter criteria	'No disaster data found for selected criteria.' Preview shows zeros	Boundary Value Analysis	Pass
TC-13	Give Feedback	Valid: All fields filled, 4-star rating, non-anonymous	Feedback saved with status 'Pending'; confirmation shown	Equivalence Partitioning	Pass
TC-14	Give Feedback	Invalid: Rating not selected (0 stars)	Validation error: 'Please rate your experience (1–5 stars).'	Boundary Value Analysis	Pass
TC-15	Give Feedback	Invalid: Feedback type = NGO Service but no entity selected	Validation error: 'Please select a specific NGO for this feedback type.'	Error Guessing	Pass
TC-16	Give Feedback	Invalid: Non-anonymous but email field left empty	Validation error: 'Please enter your email address.'	Equivalence Partitioning	Pass
TC-17	Review Feedback	Valid: Admin marks feedback as Resolved	Feedback status updated; counter updated on Admin dashboard	Equivalence Partitioning	Pass
TC-18	Review Feedback	Valid: Admin searches for feedback by keyword	Filtered list shows only matching feedback cards	Equivalence Partitioning	Pass

5.3. White-Box Testing

Method Under Test	performAssignment() — AssignRescueTeamController
Description	Assigns a rescue team to a disaster, creates a Mission, and updates team/disaster status.
Coverage Criteria	Branch Coverage
Paths/Conditions Tested	Path 1 (Happy): teamUpdated=true AND disasterUpdated=true AND missionCreated=true → shows success, reloads data. Path 2 (Team update fail): teamStatusUpdated=false → shows error, no mission created. Path 3 (Disaster update fail): assignSuccess=false → shows error, rollback attempted. Path 4 (Mission creation fail): missionCreated=false → partial success warning shown.
Coverage Result	4 of 4 branches covered (100% branch coverage)

Method Under Test	validateForm() — GiveFeedbackController
Description	Validates all required fields before feedback submission.
Coverage Criteria	Condition Coverage
Paths/Conditions Tested	Condition 1: feedbackTypeCombo.getValue() == null → returns false, shows error. Condition 2: Feedback type requires entity AND entityCombo = N/A → returns false. Condition 3: currentRating == 0 → returns false, shows star rating error. Condition 4: subjectField is empty → returns false. Condition 5: feedbackArea is empty → returns false. Condition 6: !anonymous AND email invalid → returns false. Condition 7: All conditions pass → returns true, submission proceeds.
Coverage Result	7 of 7 conditions tested (100% condition coverage)

Method Under Test	updatePreview() — GenerateReportController
Description	Fetches live data and updates the report preview panel based on selected filters.
Coverage Criteria	Statement Coverage
Paths/Conditions Tested	Statement 1: startDate or endDate is null → sets error title, resets labels, returns early. Statement 2: startDate.isAfter(endDate) → same early return path as Statement 1. Statement 3: Valid date range → fetches totals, updates all preview labels. Statement 4: selectedSeverity != 'All Severities' → appends severity tag to preview title. Statement 5: breakdown is empty → sets 'No disaster data found' in breakdown container.
Coverage Result	All 5 statement branches executed (100% statement coverage)

5.4. Bug Report Summary

Bug ID	Module	Severity	Description	Status	Fix Applied
BUG-01	Assign Rescue Team	High	Mission ID generator duplicated IDs when multiple missions existed with non-sequential IDs (e.g., MIS001, MIS010 → generated MIS002 instead of MIS011).	Fixed	Replaced manual ID generator with DB query fetching MAX ID.

BUG-02	Track Rescue Team	Medium	Map pin did not appear when team's city was not found in MAPLOCATION table; NullPointerException thrown.	Fixed	Added null check in updatePinPosition(); falls back to default map center coordinates.
BUG-03	Manage Disasters	High	The status of rescue and ngo team didn't update when the disaster was deleted.	Fixed	Added team id fetcher and updated teams' statuses, if they were assigned to the respective disaster..
BUG-04	Review Feedback	Low	The color of labels on feedback differed from the buttons of labels.	Fixed	Swapped the colors of labels.
BUG-05	Review Feedback	Low	Average rating label on Admin Dashboard showed 'NaN/5.0' when no feedback existed in the database.	Fixed	Added null/NaN guard in loadFeedbackStats(); defaults to 0.0 when no ratings exist.

5.5. Test Results and Observations

Total test cases executed: 18

Passed: 18 | Failed: 0 | Blocked: 0

Key observations:

- All critical user flows (assign NGO, assign rescue team, mission status update) pass with correct database state transitions.
- Report generation correctly handles boundary cases such as equal start/end dates and empty result sets.
- Feedback validation correctly blocks submission for all identified invalid input combinations.
- Five bugs were identified during testing; all five were resolved before sprint completion.
- White-box analysis confirmed 100% branch, condition, and statement coverage on the three critical methods tested.

How testing improved system quality:

Testing in Sprint 3 specifically uncovered the Mission ID duplication bug, which could have caused data integrity failures in production. The null-pointer exception in the Track Rescue Teams map view was also caught before delivery, preventing a runtime crash. Systematic condition coverage of the feedback validator eliminated all edge-case submission bypasses. Overall, the test activities increased confidence in the correctness and robustness of the final integrated system.